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# Titanic Survival Prediction - NeuroNexus Task
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# Dataset: tested.csv (uploaded)

# -----
# 1. Import Required Libraries
# -----
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

import warnings
warnings.filterwarnings('ignore')

# -----
# 2. Load the Dataset
# -----
df = pd.read_csv("tested.csv")
df.head()

# -----
# 3. Initial Data Inspection
# -----
print(df.info())
print("\nMissing values:\n", df.isnull().sum())

# -----
# 4. Data Cleaning
# -----

# Fill missing Age with median
df['Age'].fillna(df['Age'].median(), inplace=True)

# Fill missing Embarked with most frequent
df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)

# Drop Cabin (too many missing values)
df.drop(['Cabin', 'Name', 'Ticket', 'PassengerId'], axis=1, inplace=True)

# Encode categorical features
label_enc = LabelEncoder()
df['Sex'] = label_enc.fit_transform(df['Sex'])
df['Embarked'] = label_enc.fit_transform(df['Embarked'])

df.head()

# -----
# 5. Exploratory Data Analysis
# -----

# Survival rate by gender
sns.countplot(data=df, x='Survived', hue='Sex')
plt.title('Survival Rate by Gender')
plt.show()

# Survival rate by class
sns.countplot(data=df, x='Survived', hue='Pclass')
plt.title('Survival Rate by Passenger Class')
plt.show()

# Heatmap of correlations
plt.figure(figsize=(10, 6))
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
plt.title('Correlation Matrix')
plt.show()

# -----
# 6. Train/Test Split
# -----
X = df.drop('Survived', axis=1)
y = df['Survived']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# -----
# 7. Model Training
# -----
model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)
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y_pred = model.predict(X_test)
# -----
# 8. Model Evaluation
# -----
print("Accuracy Score:", accuracy_score(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))

# Confusion Matrix
conf_matrix = confusion_matrix(y_test, y_pred)
sns.heatmap(conf_matrix, annot=True, fmt='d', cmap='Blues')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix')
plt.show()
```

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<class 'pandas.core.frame.DataFrame'>
RangeIndex: 418 entries, 0 to 417
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   PassengerId  418 non-null    int64
1   Survived     418 non-null    int64
2   Pclass       418 non-null    int64
3   Name         418 non-null    object
4   Sex          418 non-null    object
5   Age          332 non-null    float64
6   SibSp        418 non-null    int64
7   Parch        418 non-null    int64
8   Ticket       418 non-null    object
9   Fare         417 non-null    float64
10  Cabin        91 non-null     object
11  Embarked     418 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 39.3+ KB
None

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Missing values:
 PassengerId    0
 Survived       0
 Pclass         0
 Name           0
 Sex            0
 Age           86
 SibSp          0
 Parch          0
 Ticket         0
 Fare           1
 Cabin         327
 Embarked       0
dtype: int64

```

